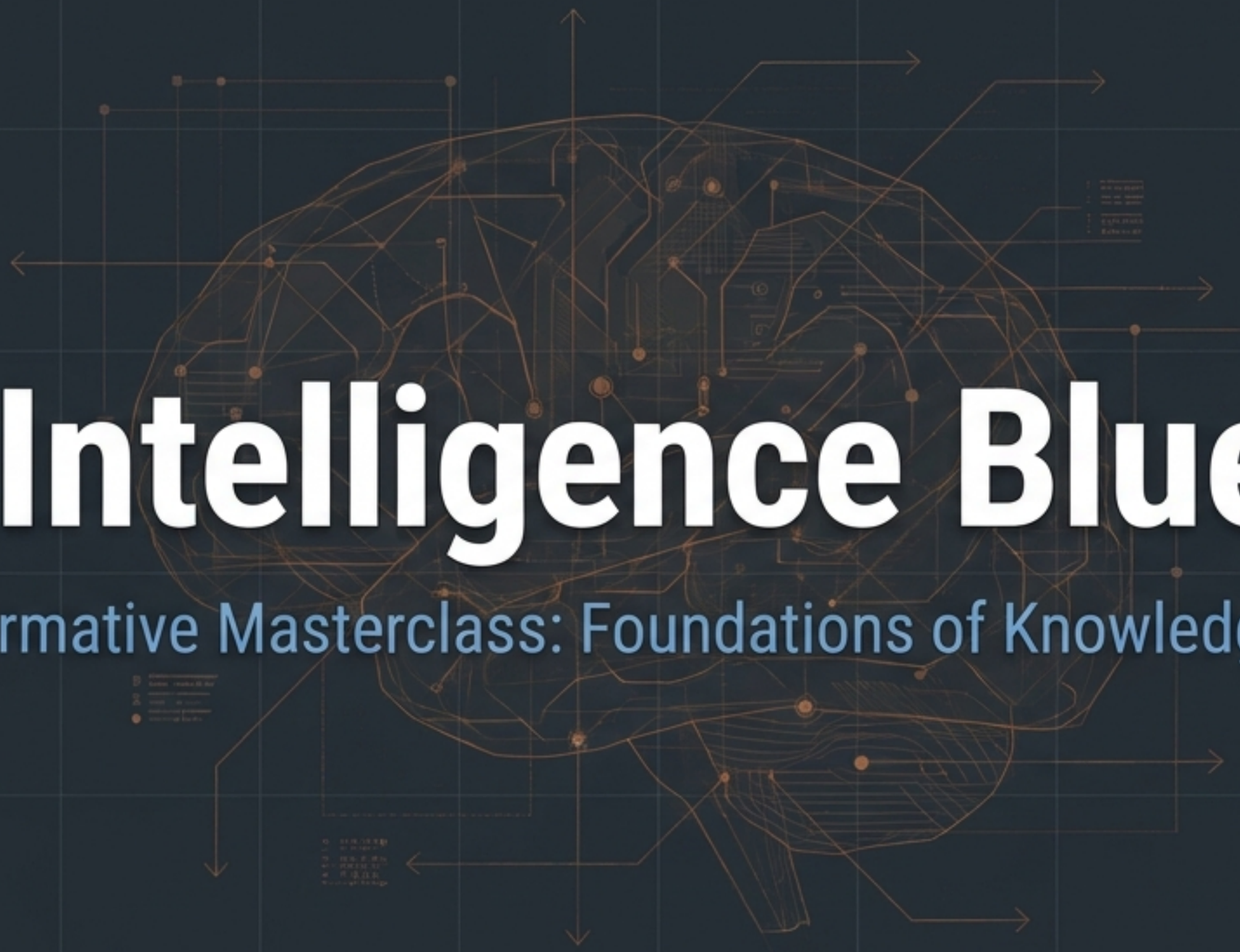


The Intelligence Blueprint

COM759 Formative Masterclass: Foundations of Knowledge Engineering

Initialising Diagnostic Terminal... [15 Core Challenges Loaded]

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THE TERMINAL

Q1. According to the modern definition of Artificial Intelligence (as aligned with Russell & Norvig), how is an ‘intelligent agent’ primarily defined in the context of computer science?

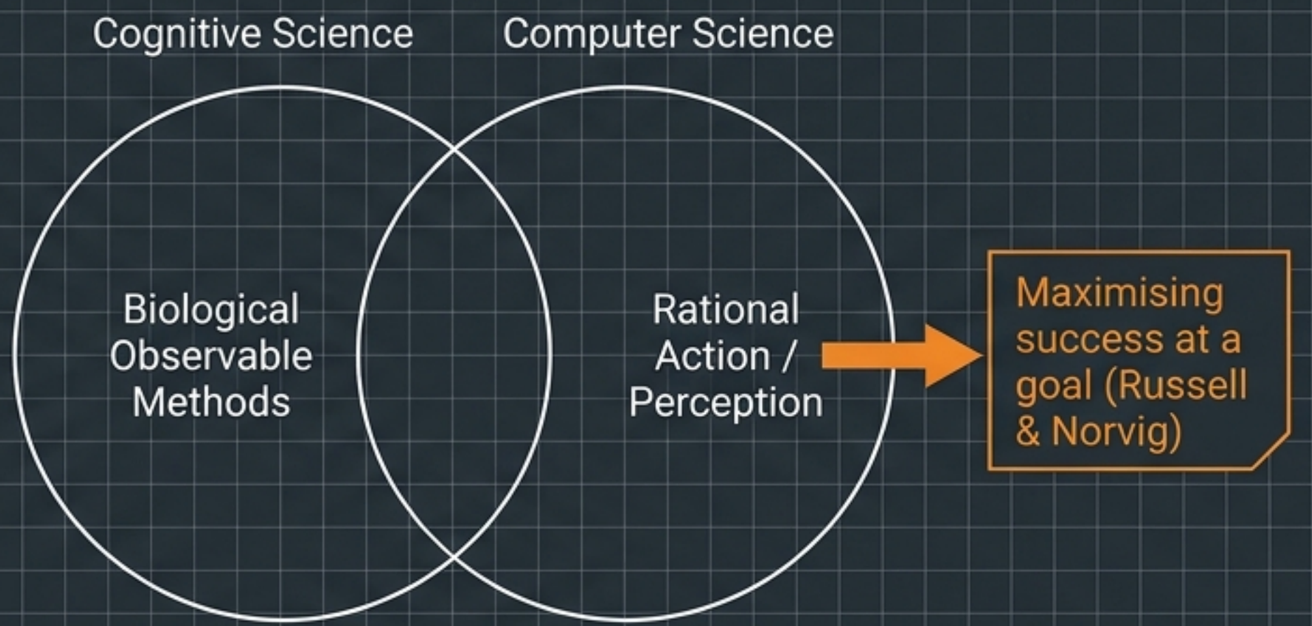
A) By its ability to perfectly replicate biologically observable human thought processes.

B) By its capacity to perceive its environment and take actions that maximise its chance of success at some goal.

C) By its success in passing the Turing Test against expert interrogators.

D) By its utilisation of massive datasets and cloud computing environments.

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Q2. Which of the following statements most accurately reflects the current state of Artificial General Intelligence (AGI) as discussed in the Week 1 foundations?

A) We are rapidly approaching AGI due to the 175 billion parameters of models like ChatGPT-3.

B) AGI is currently achieved through massive combinatorial search algorithms.

C) There is no empirical evidence that we are close to Strong AI; current dominance belongs entirely to pattern-based Weak AI.

D) Strong AI is defined by a system's ability to solve specific domain problems without human intervention.

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Weak / Narrow AI	Strong AI / AGI
	
Dominant Mode Today	No Current Empirical Evidence
Pattern Detection	Abstract Reasoning

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Q3. In the First Generation of AI (Symbolism), the semantics of symbols are described as “parasitic.” What does this mean in a Knowledge Engineering context?

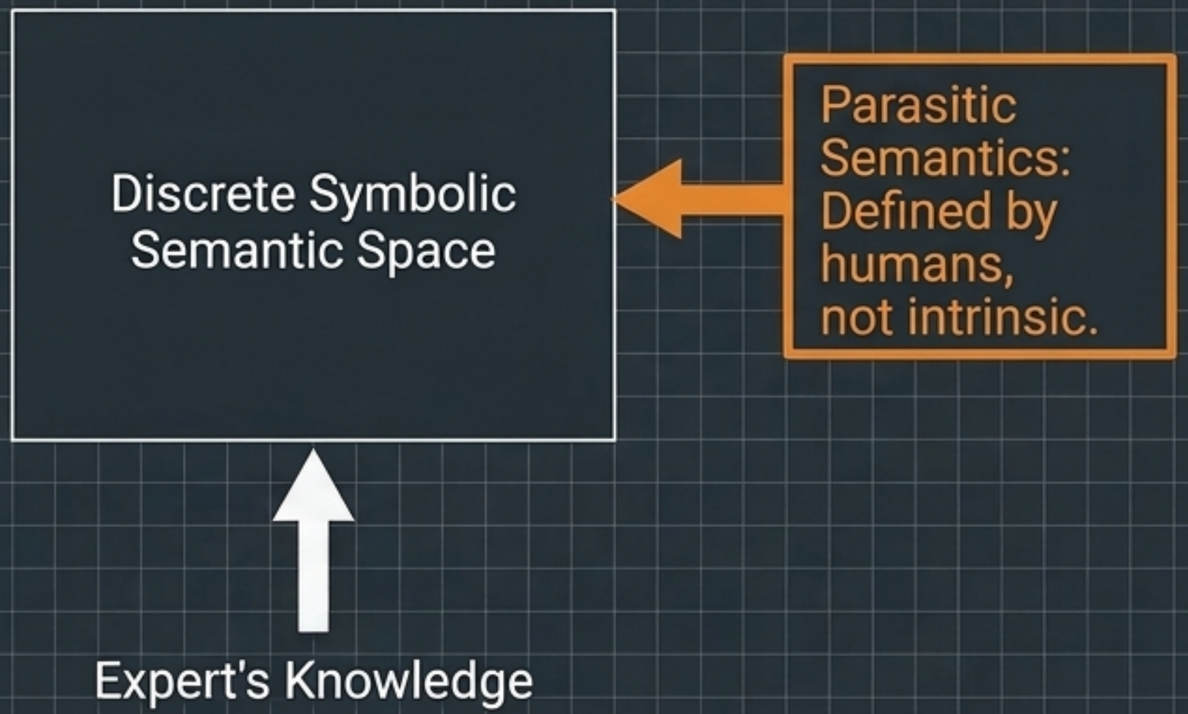
A) The symbols extract computational power from other running programmes.

B) The meaning of the symbols is intrinsically generated by the machine's learning algorithm.

C) The machine requires continuous data feeds from external APIs to function.

D) The semantics of the symbols are defined entirely by humans, rather than being intrinsically understood by the machine.

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Q4. What represents the fundamental shift in the basic unit of cognition when moving from First Generation (Symbolic) AI to Second Generation (Connectionism) AI?

A) Moving from rules-based algorithms to probabilistic algorithms like Hidden Markov Models.

B) Moving from explicit human-defined symbols to sub-symbols (vectors) operating in a continual data space.

C) Moving from supervised learning to purely unsupervised reinforcement learning.

D) Moving from CPU-based processing to GPU-based parallel processing.

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Q5. IBM's Deep Blue (1997) successfully conquered the game of chess. Which combination of factors was strictly responsible for this milestone?

A) 30M Go positions, Monte Carlo tree search, and 280 GPUs.

B) **700K human-played positions**, α-β pruning algorithms, and 200M moves/sec computational capacity.

C) Unsupervised learning, qualitative reasoning, and cloud-based big data environments.

D) 175 billion parameters, deep neural networks, and 800 petaflops of processing power.

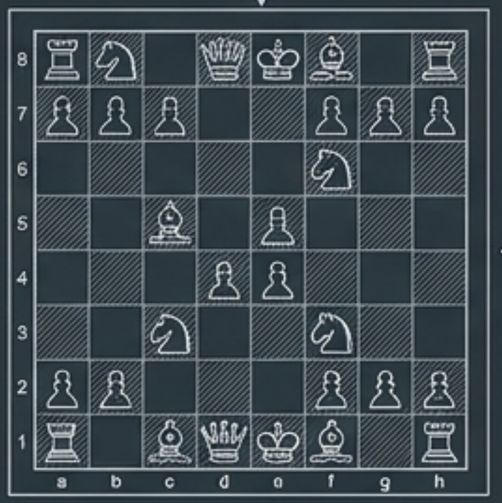
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Equation of Success

KNOWLEDGE
700K human positions

ALGORITHM
α-β pruning

COMPUTE
IBM RS/6000 SP2, 200M moves/sec



3×10^{120}
Complexity

PARADIGM: RATIONAL REASONING

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Q6. Compared to Deep Blue's approach to chess, how did Google DeepMind's AlphaGo address the vastly larger combinatorial explosion (2×10^{170}) of Go in 2016?

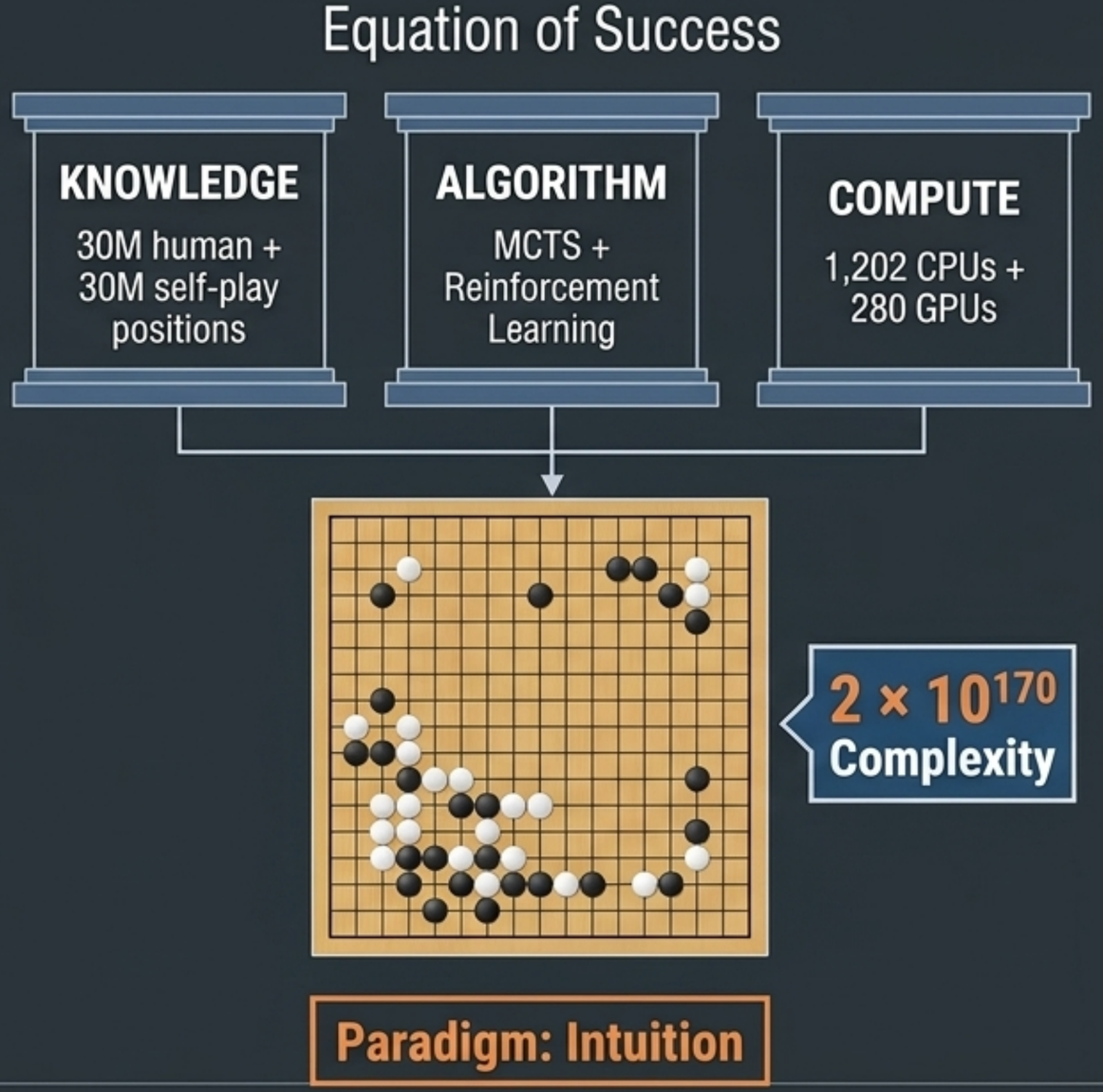
A) By utilising an infinitely scalable quantum computing architecture.

B) By manually coding a more complex set of domain-specific heuristics and α-β pruning rules.

C) By replacing traditional search entirely with a single massive deep neural network.

D) By combining Monte Carlo Tree Search with Deep Reinforcement Learning (self-play) to mimic "intuition".

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Q7. The training of ChatGPT-3 illustrates the massive resource dependency of modern deep learning. Which of the following accurately reflects the scale of this model?

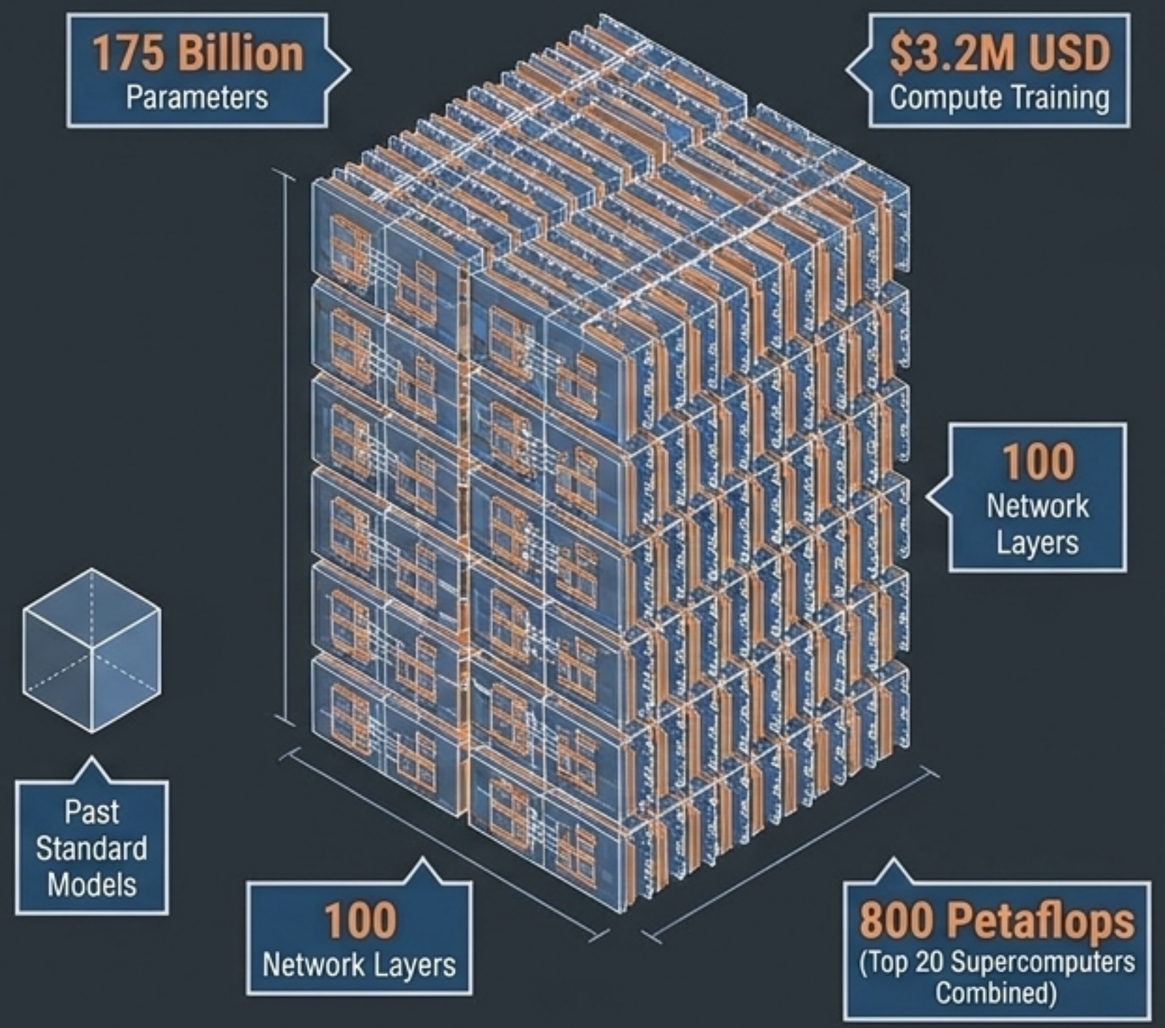
A) 1 million parameters, trained on standard consumer GPUs.

B) 175 billion parameters, requiring roughly 800 petaflops of processing power (equivalent to 20 supercomputers).

C) 175 million parameters, requiring $3.2 billion in computational resources.

D) A completely decentralised architecture requiring no centralised cloud computing.

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Q8. According to the performance curves comparing traditional learning models to deep learning networks, what happens as dataset volume transitions into the "Big Data" era?

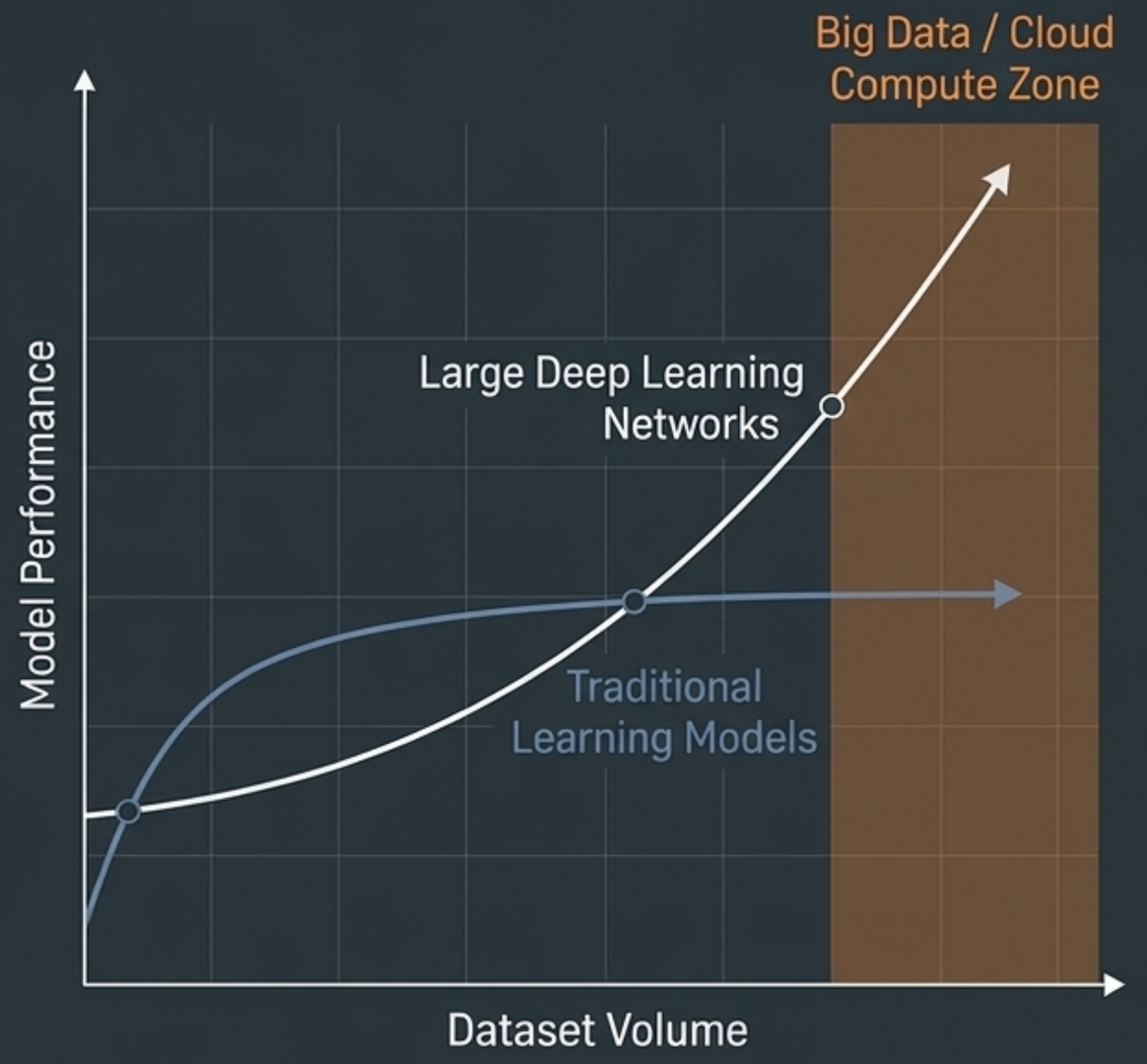
A) Traditional models maintain parallel performance curves to deep learning models.

B) Large deep learning networks suffer from immediate overfitting and decline in accuracy.

C) Traditional learning models plateau in performance, while large deep learning networks continue to scale in accuracy.

D) Manual feature engineering becomes increasingly necessary for large deep learning networks to function.

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Q9. Despite their high accuracy, data-driven Deep Learning models exhibit critical weaknesses. Which of the following best describes the phenomenon of **“non-robustness”** in these models?

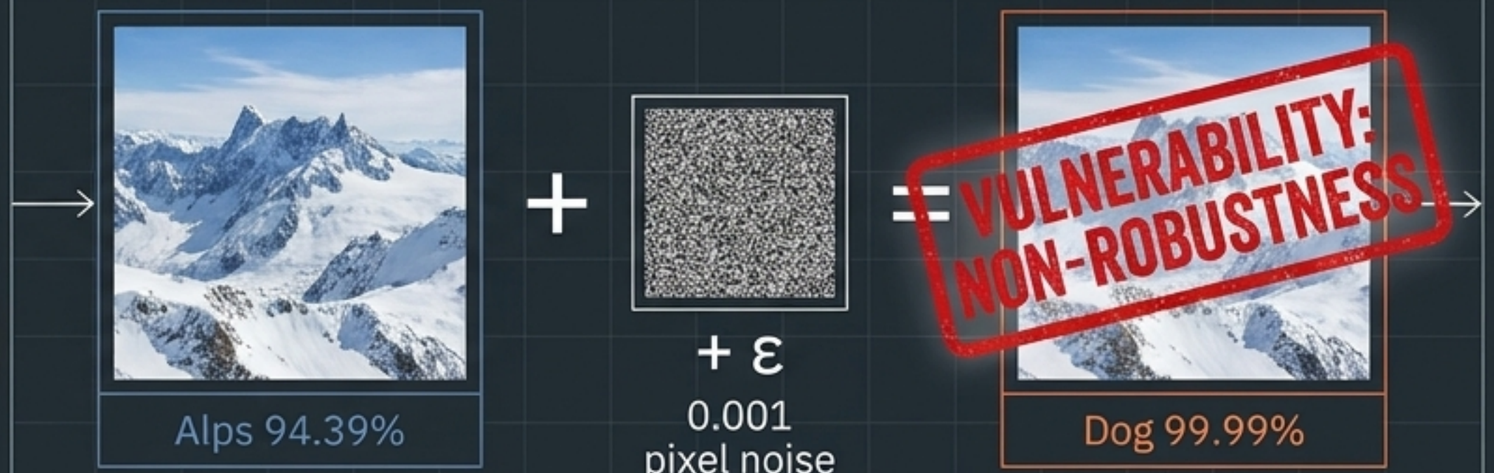
A) The inability to process more than 5 dimensions of data simultaneously.

B) The introduction of imperceptible mathematical noise (ε) can cause the model to confidently misclassify an image (e.g., classifying a mountain as a dog).

C) The models frequently delete their own training data to save memory.

D) The requirement for human experts to manually tag every pixel in an image.

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Q10. In the formal study of Logic as a method for Knowledge Representation and Reasoning (KR&R), what role does "Proof Theory" play?

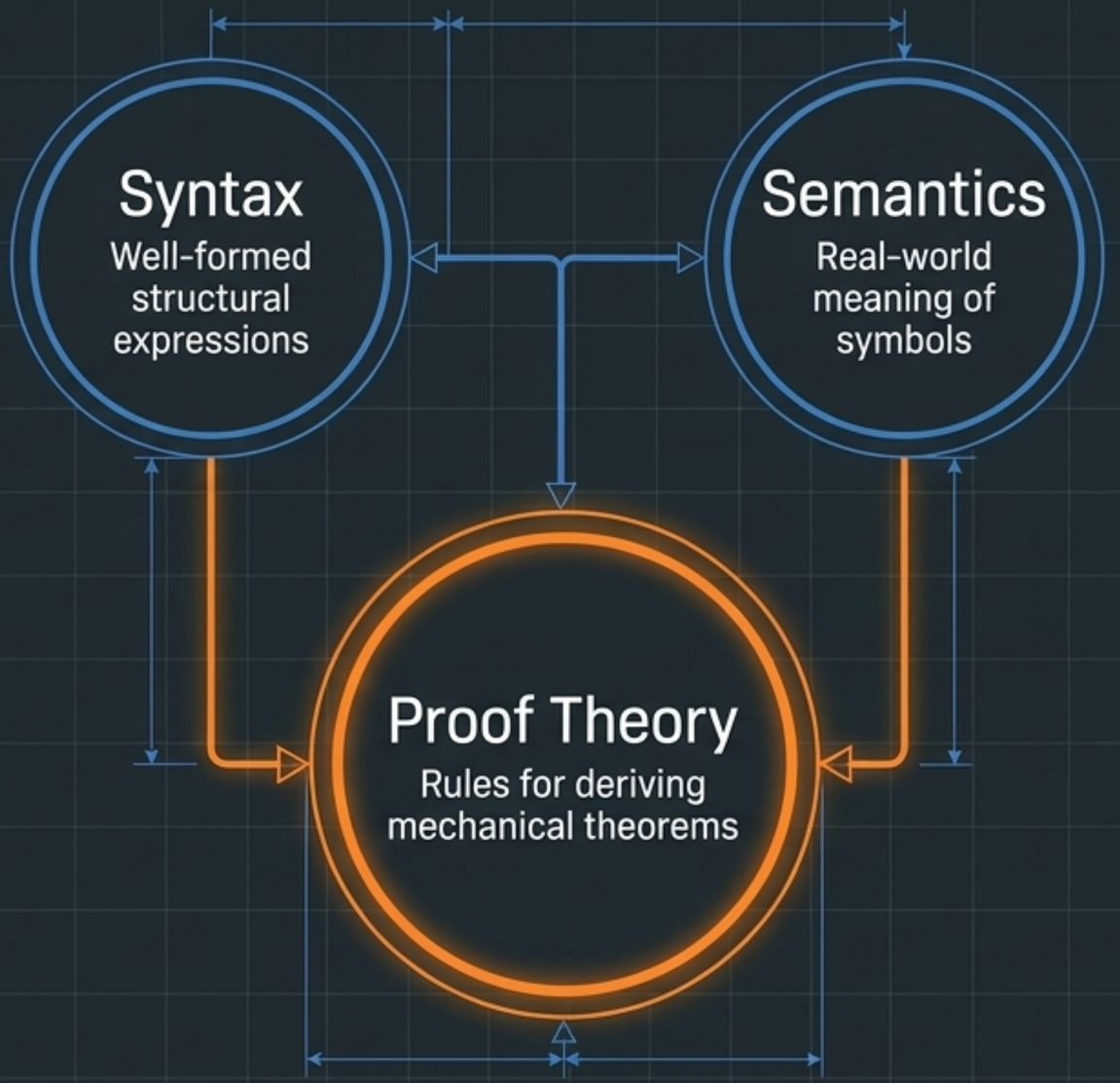
A) It defines the atomic symbols of the logical language.

B) It determines the real-world truth and meaning of the atomic symbols.

C) It provides the rules for determining a subset of logical expressions called theorems, allowing the derivation of inferred statements.

D) It strictly limits the computational capacity of the knowledge base.

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Q11. In Logic Programming (such as Prolog), how do core Artificial Intelligence components map to formal Logic components?

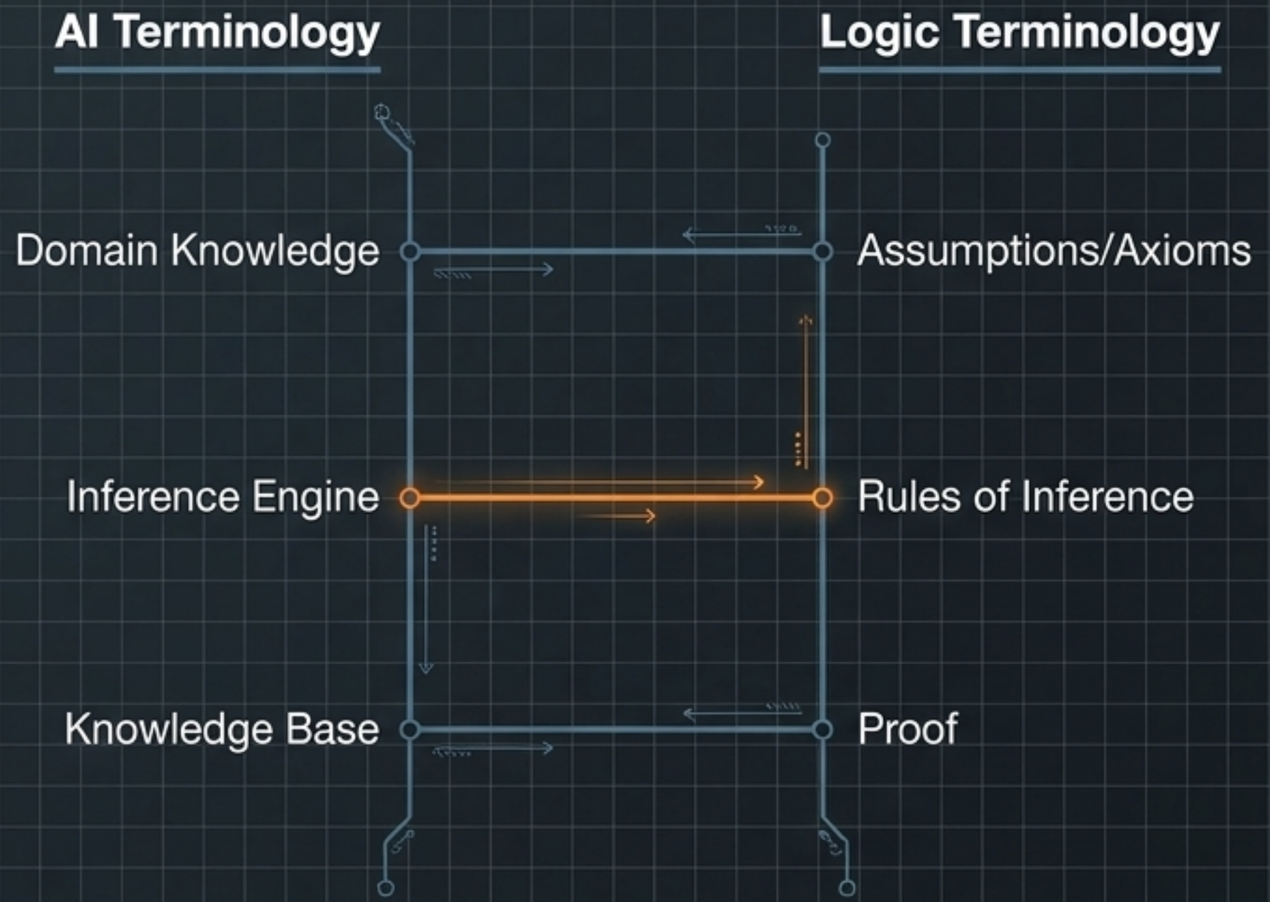
A) The “Inference Engine” maps to “Assumptions/Axioms”.

B) “Domain Knowledge or Rules” maps to “Rules of Inference”.

C) The “Inference Engine procedures” map to the “Rules of Inference”.

D) The “Knowledge Base” maps directly to “Syntax”.

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Q12. What is the defining characteristic of a “non-monotonic” reasoning logic?

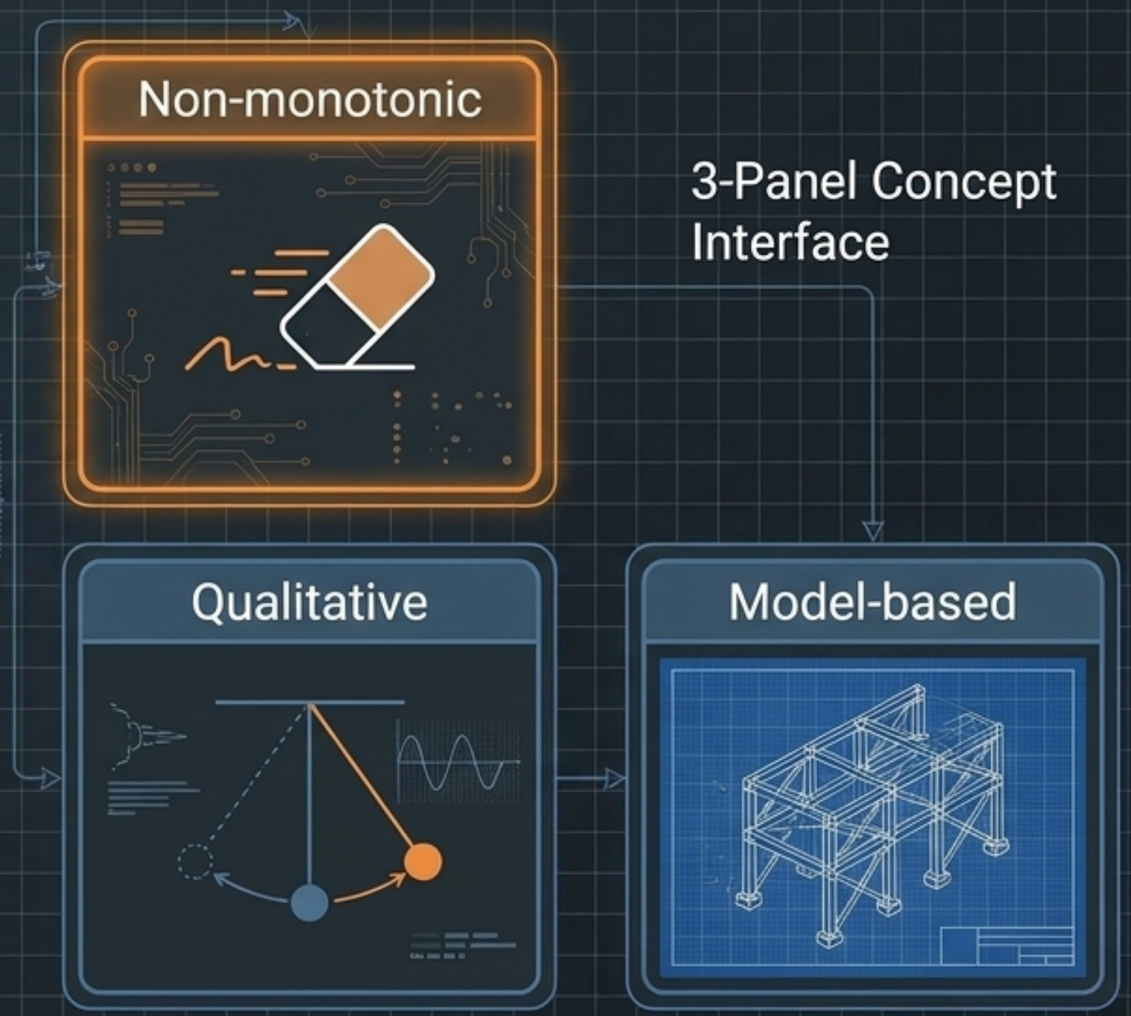
A) It reasons strictly with continuous, quantitative data rather than discrete symbols.

B) Some previous conclusions can be invalidated by adding new knowledge to the system.

C) It generates completely novel ideas and is associated purely with inductive creativity.

D) It holds that a statement (A) is necessarily true at all points in time (temporal logic).

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Q13. According to DARPA's framework for Explainable AI (XAI), what is the primary limitation of the current generation of Machine Learning applications?

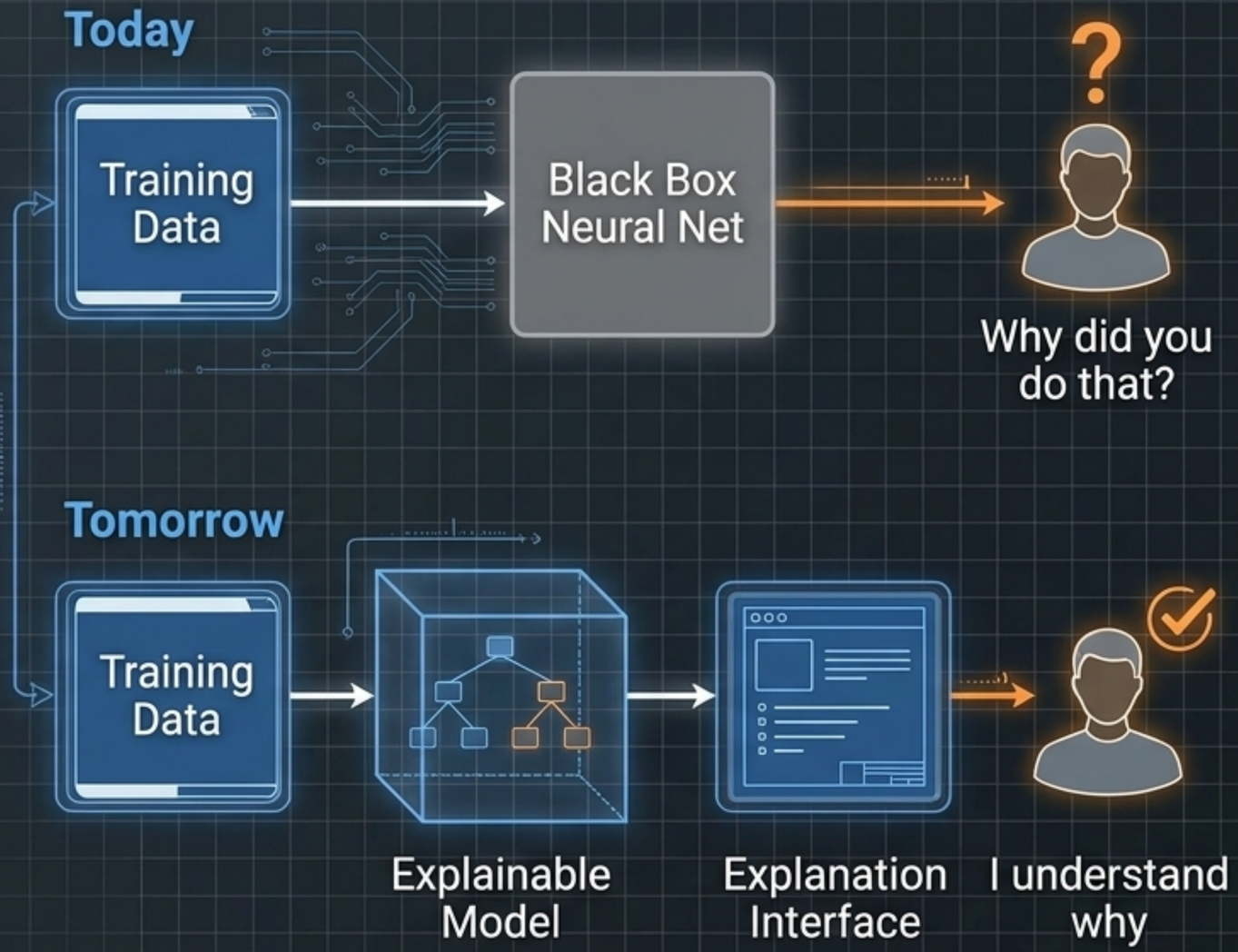
A) They do not possess enough computational power to process real-time video.

B) They are incapable of being trained on diverse, unstructured datasets.

C) Their effectiveness is limited by their opaque, non-intuitive nature, preventing users from appropriately trusting and managing them.

D) They output logic proofs that are too mathematically complex for non-specialists to read.

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Q14. When designing AI systems, researchers often face a trade-off between different architectures. Generally, how do Deep Neural Networks compare to traditional Expert Systems in this trade-off?

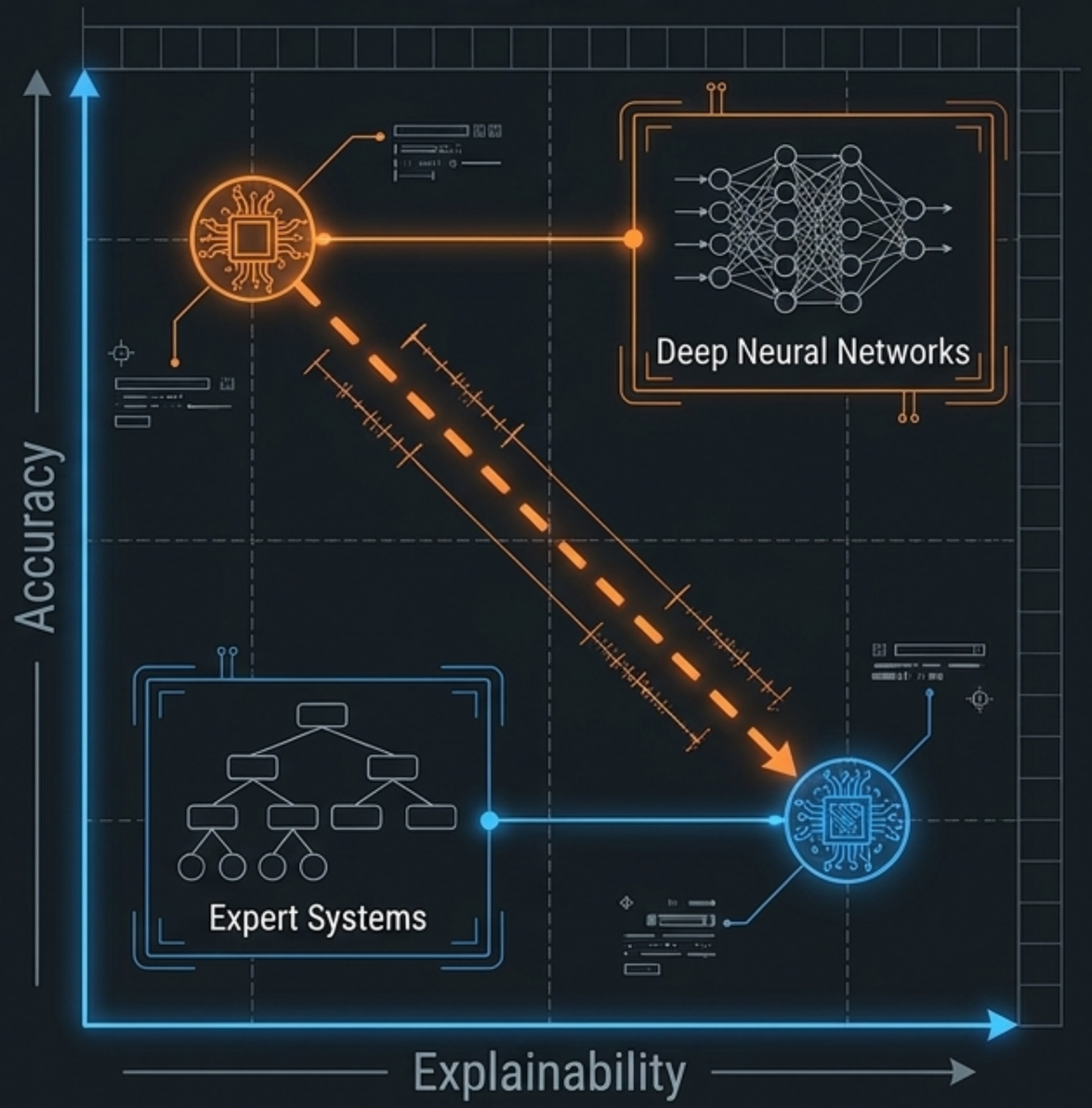
A) Neural networks offer excellent explanations but suffer from poor accuracy.

B) Expert systems offer both higher accuracy and better explanations than neural networks.

C) Neural networks are highly accurate but provide poor explanations, whereas expert systems provide good explanations but generally lower accuracy on complex data.

D) There is no trade-off; both systems offer identical explainability.

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Q15. Under the EU General Data Protection Regulations (GDPR) framework for Trustworthy AI, what specific right is granted to users regarding algorithms?

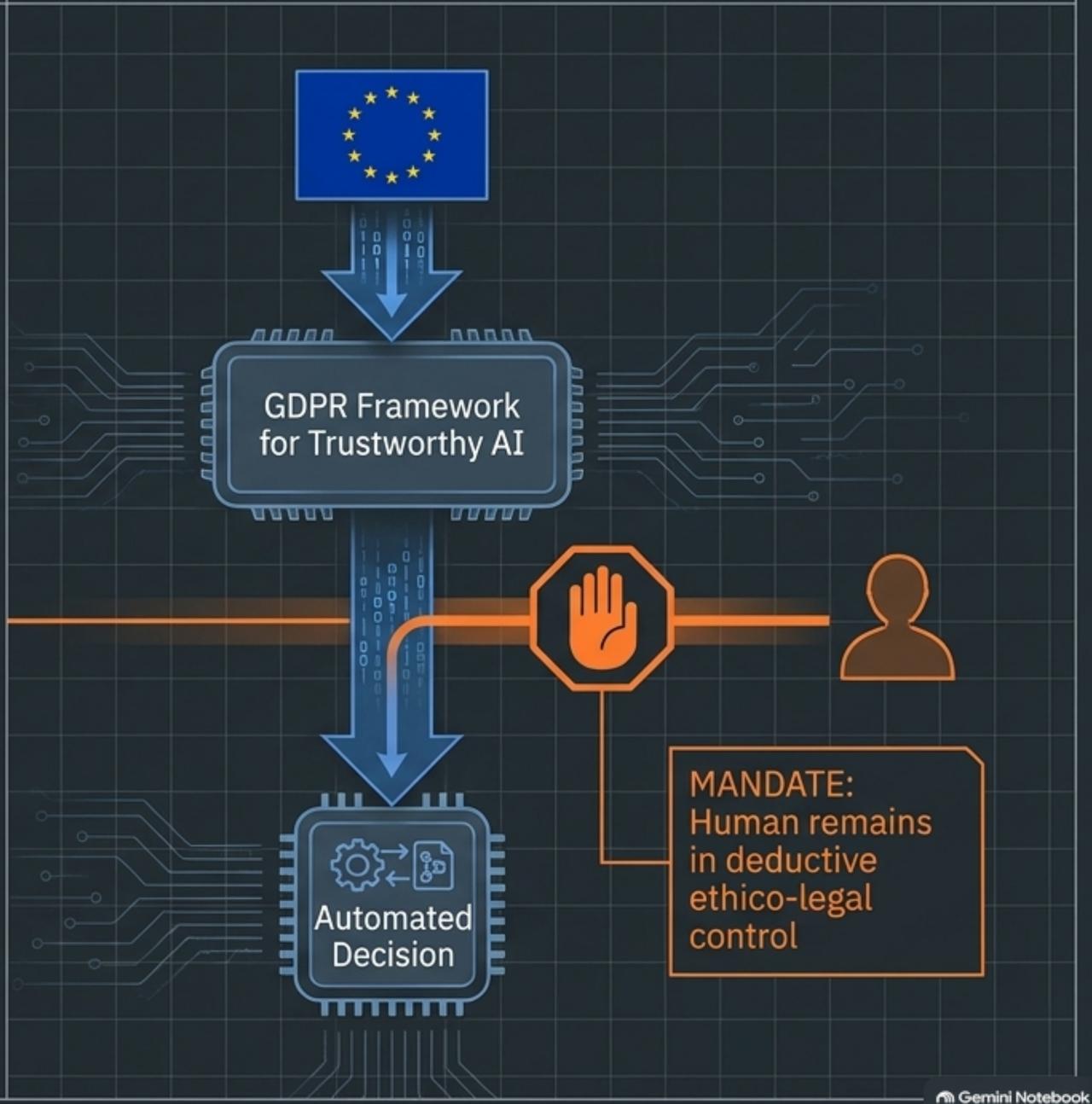
A) The right to demand the proprietary source code of any commercial AI system.

B) The right to contest a completely automated decision if it has legal or other significant effects on them.

C) The right to delete their personal data from the hidden layers of a trained neural network.

D) The right to receive financial compensation every time an AI uses their data for training.

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